

ADI KARTHIKEYA SUNKARA BALASUBRAMANYAM

Houston, TX | +1 832 654 1709 | adikarthikeya1234@gmail.com

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EXPERIENCE

Research Assistant, Machine Learning & AI <i>University of Houston (Petroleum Industry Partnership)</i>	Jan 2025 – Present Houston, TX
- Architected and deployed end-to-end ML models for subsurface analytics and resource optimization, leveraging Databricks and SQL Server for large-scale data processing. - Automated data validation and model evaluation pipelines using Python, standardizing output evaluation and reducing turnaround time by 40% across concurrent analytical workflows. - Designed scalable RAG architectures and vector databases (ChromaDB) to query technical documentation, orchestrating agentic workflows via FastAPI and LLMs.	
Data Science Intern, Geospatial ML <i>NSF Data Science for Energy Transition, University of Houston</i>	May 2025 – Jun 2025 Houston, TX
- Trained classification pipeline (PCA, Random Forest, XGBoost) on 50,000+ geological survey records; reached 80.24% accuracy, a 7% lift over the logistic regression baseline. - Built interactive Tableau and QGIS dashboards for NSF stakeholders showing model predictions on candidate energy-transition sites; rewrote slow joins in Python (pandas, NumPy) cutting pipeline runtime by 35%.	
Data Analyst, ETL and BI <i>Corizo Edutech</i>	Apr 2023 – May 2024 Bengaluru, India
- Built Python and SQL ETL pipelines moving 1M+ records from internal sources into Azure Data Lake and downstream analytics warehouses; cut runtime by 60% and stabilized reporting. - Trained XGBoost and Random Forest models for customer segmentation; reached 89% accuracy and 0.91 ROC-AUC, a 12% lift over the prior baseline used in production targeting. - Shipped Power BI dashboards covering 8 KPIs across 3 business units; adopted weekly by leadership for ops reviews and quarterly business reporting.	

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, MySQL, SQL Server, BigQuery), R
ML & AI: PyTorch, TensorFlow, HuggingFace, scikit-learn, XGBoost, LightGBM, SHAP, MLflow, LangChain, LangGraph, Google ADK, RAG, ChromaDB, Neo4j, SBERT, Gemini, Claude, OpenAI APIs
Cloud & MLOps: Azure Data Lake, Databricks, GCP (Cloud Run, BigQuery, Vertex AI), AWS, Docker, CI/CD, Git, GitHub, Linux
Backend & Visualization: FastAPI, Flask, REST APIs, Streamlit, Power BI, Tableau, Pandas, NumPy

PROJECTS

RareDx, Agentic Rare Disease Diagnostic Assistant <i>LangGraph, Neo4j, ChromaDB, BioLORD</i>	2026
- Rare disease diagnostics often suffer from low accuracy and prolonged timelines due to fragmented and dense medical literature. - Built agentic diagnostic workflow in LangGraph combining Neo4j graph retrieval with ChromaDB semantic search over Orphanet disease profiles; lifted Top-1 accuracy from 10% to 92.5% and Recall@5 from 22.5% to 100% on a 40-case benchmark. - Ran ablations across 7 retrieval strategies (280 evaluations); calibrated diagnostic confidence (Brier 0.068, ECE 0.051) and enforced evidence-only outputs via Neo4j paths and PubMed citation grounding for 0% hallucination rate.	
ShopperAI, Agentic AI Shopping Assistant <i>Google ADK, Gemini, GCP Cloud Run, FastAPI, Docker</i>	2026
- E-commerce shoppers struggle to find specific, highly constrained items through traditional keyword search, leading to high cart abandonment rates. - Ranked 4th globally out of 200 teams at ODSC AI x Google Cloud Agentic AI Hackathon; built production agentic shopping assistant with deterministic stage-based workflows deployed on GCP Cloud Run. - Reduced hallucinated outputs and improved response relevance through structured prompt engineering, tool-use constraints, and grounded retrieval inside the Google ADK framework.	
RAG-Powered Document Intelligence System <i>LangChain, ChromaDB, SBERT, FastAPI, GCP</i>	2024
- Enterprise teams waste thousands of hours manually searching through dense, unstructured corporate documentation for operational answers. - Built RAG pipeline over 1,000+ documents using SBERT and ChromaDB; reached 92% retrieval accuracy and 18% output quality gain across automated RAG evaluation metrics. - Deployed on GCP with async FastAPI endpoints sustaining sub-500ms latency and 99.5% uptime under load testing.	

EDUCATION

University of Houston , M.S. Data Science, GPA 3.89/4.00	Houston, TX	May 2026
Amrita Vishwa Vidyapeetham , B.Tech, ECE, GPA 3.50/4.00	Bengaluru, India	May 2024